



Contribution to the Supplement: 'Lobsters in a Changing Climate' Original Article

Aggressive behaviour of spotted spiny lobsters (*Panulirus guttatus*) in different social contexts: the influence of sex, size, and missing limbs

Patricia Briones-Fourzán*, Roberto Domínguez-Gallegos, and Enrique Lozano-Álvarez

Instituto de Ciencias del Mar y Limnología, Unidad Académica de Sistemas Arrecifales, Universidad Nacional Autónoma de México, Puerto Morelos, Quintana Roo 77580, Mexico

*Corresponding author: tel: +52 998 8710009; fax: +52 998 8710138; e-mail: briones@cmarl.unam.mx

Briones-Fourzán, P., Domínguez-Gallegos, R., and Lozano-Álvarez, E. Aggressive behaviour of spotted spiny lobsters (*Panulirus guttatus*) in different social contexts: the influence of sex, size, and missing limbs. – ICES Journal of Marine Science, 72: i155–i163.

Received 14 August 2014; revised 31 October 2014; accepted 3 November 2014; advance access publication 26 November 2014.

Panulirus guttatus is a sedentary spiny lobster that exhibits cryptic behaviour and a low degree of gregariousness. Because these lobsters are obligate coral reef-dwellers and avoid sandy expanses, they are potentially distributed in relatively small, discrete populations with variable social contexts, which can strongly influence the expression of aggression. The present study examined the relative importance of sex, size, and the number of missing limbs in the shelter-related aggressive behaviour of replicated groups of four lobsters that differed in social context (i.e. same-sex and mixed-sex groups). Each group was held in a seawater tank with a single artificial cave-like shelter. The interior of the cave was video-recorded for 72 h and the number of aggressions performed by each individual was quantified in a 10-min segment of video per hour. Most aggressions were related to occupancy of the shelter inner space and tended to end when individuals were sufficiently spaced out. In general, per-capita rates of aggression were higher at night and size was an important predictor of aggressiveness among individuals of the same sex. Males were substantially more aggressive than females, but the number of missing limbs significantly impacted the degree of aggressiveness in males. In mixed-sex groups, fewer aggressions occurred when the largest individual was a male than when it was a female, suggesting that it may take longer for smaller males to assert themselves. Our results provide insights into some potential consequences of increase in fishing pressure and loss of habitat complexity in Caribbean reefs for the social behaviour and population dynamics of these lobsters.

Keywords: aggression, agonistic interactions, Caribbean coral reefs, social behaviour, social context, spiny lobsters.

Introduction

In many animal systems, aggressive behaviour is an important component of agonistic and social behaviour that includes actual attacks or any threats of attack (Huber and Kravitz, 2010). Although aggression and dominance are not necessarily correlated (Hand, 1986; Drews, 1993), large crustacean males are typically more aggressive and tend to dominate smaller conspecifics, often monopolizing resources such as food, mates, or shelter (Lee and Fielder, 1983; Thorpe *et al.*, 1994; Edsman and Jonsson, 1996; Beattie *et al.*, 2012). In clawed decapods (e.g. crabs, crayfish, and clawed lobsters), the claws constitute potentially deadly weapons and hence agonistic interactions involve ritualized displays that serve as tests of strength rather than actual attacks (Neil, 1985;

Huber and Kravitz, 1995; Davis and Huber, 2007). In contrast, in the clawless spiny lobsters (Decapoda: Achelata: Palinuridae), ritualized threat behaviours have not evolved and agonistic behaviour can involve a great deal of contact and struggle, but is less likely to result in injury or death (Atema and Cobb, 1980; Cobb, 1981).

Spiny lobsters are large, nocturnally active crustaceans that depend on available structured shelter to hide from predators. Spiny lobsters exhibit gregarious sheltering, a behaviour mediated by conspecific chemical communication (Childress, 2007). Although the degree of gregariousness varies with species, even in highly gregarious species individuals can behave aggressively towards conspecifics attempting to share shelters and may form dominance hierarchies

that determine their position within the shelter (Fielder, 1965; Berrill, 1976; McKoy, 1979; Cobb, 1981; Shabani *et al.*, 2009).

Panulirus guttatus, the Caribbean spotted spiny lobster, is a relatively small, coral reef-obligate palinurid (Sharp *et al.*, 1997). Individuals of *P. guttatus* have a limited home range (Lozano-Álvarez *et al.*, 2002; Negrete-Soto *et al.*, 2002) and hide deeply in reef crevices from which they emerge only for brief periods at night to forage on the reef itself (Wynne and Côté, 2007). Although *P. guttatus* lobsters are attracted to conspecific odours, especially when reproductive activity is high (Briones-Fourzán and Lozano-Álvarez, 2005), they exhibit a low propensity to aggregate, and when multiple individuals co-occupy a reef crevice, they tend to space out in separate cavities or locations within the crevice (Childress, 2007; Briones-Fourzán *et al.*, 2013). Similarly, in a laboratory setting using artificial cave-like shelters, large males aggressively displaced smaller individuals or evicted them altogether from the cave, but aggressions subsided once each lobster secured a separate location within the cave (Lozano-Álvarez and Briones-Fourzán, 2001; Segura-García *et al.*, 2004). This spacing behaviour of *P. guttatus* contrasts with other spiny lobsters, wherein individuals sharing crevices typically maintain close contact with each other. Therefore, the distribution of *P. guttatus* lobsters over the reef habitat is modulated by a complex interplay between chemical attraction and aggressive behaviour (Briones-Fourzán and Lozano-Álvarez, 2005).

Although large males of *P. guttatus* are generally more aggressive than smaller conspecifics (Lozano-Álvarez and Briones-Fourzán, 2001; Segura-García *et al.*, 2004), this does not imply that large males should dominate all agonistic encounters because all individuals, irrespective of sex and size, have the potential to express aggression and the observed aggressiveness is a circumstantial product of the cost/benefit ratio of using it (Drews, 1993; Huber and Kravitz, 2010). Moreover, aggressiveness can be strongly influenced by social context (Karnofsky *et al.*, 1989; Graham and Herberholz, 2009; Beattie *et al.*, 2012), which is particularly relevant to *P. guttatus* because these lobsters are restricted to a highly patchy habitat and hence potentially distributed in fragmented populations with variable social contexts (Sharp *et al.* 1997; Negrete-Soto *et al.*, 2002; Robertson and Butler, 2009; Briones-Fourzán *et al.*, 2013). Given the great amount of time that *P. guttatus* lobsters spend sheltered, we examined in the laboratory the influence of social context (i.e. same-sex and mixed-sex groups) on their shelter-related aggressive behaviour. In particular, we were interested in determining the effect of social context on the rate of aggression, whether individual traits affect aggressiveness of lobsters in a given social context, and whether there is a significantly more aggressive individual in every lobster group irrespective of social context. Following the observations of Lozano-Álvarez and Briones-Fourzán (2001) and Segura-García *et al.* (2004), we predicted that the rate of aggression would be significantly affected by social context, with males exhibiting higher levels of aggressiveness than females, but that the largest individuals would exhibit a higher degree of aggressiveness regardless of social context.

Material and methods

Lobster collection and holding

Lobsters were collected from the Puerto Morelos coral reef (eastern margin of the Yucatan peninsula, Mexico), put into a seawater container in a boat and transported to our laboratory, located at Puerto Morelos (20°52.1'N, 86°52.1'W), within an hour of collection. Lobsters were allowed to acclimate for up to 8 d in large

tanks fitted with plenty of individual shelters to reduce the probability of shelter-related fights. The shelters were segments of PVC pipe sealed on one end, both to make them darker and to allow the lobsters to fully retreat into them (Briones-Fourzán and Lozano-Álvarez, 2005). The holding tanks (and the experimental units mentioned below) received seawater from the Puerto Morelos reef lagoon via an open-flow system and were in the open but under a translucent shade and hence subject to ambient light. During the holding period, lobsters were fed *ad libitum* every other day with frozen crabs and mussels, previously thawed. All experimental lobsters were adults.

Experimental design

The experimental units were three fibreglass tanks (2 m in diameter and 0.9 m in depth, with ~2500 l of seawater), each containing a single artificial shelter in the form of a cave with two low entrances. The cave was constructed with an inverted plastic trash bin measuring 0.45 m in diameter and 0.75 m in height with the rim weighed to increase stability. Because *P. guttatus* lobsters tend to cling to the walls or ceiling of reef crevices, the interior of the cave was lined with wire mesh slightly separated from the walls to provide gripping surface for the lobsters. The mesh reduced the inner height of the cave to ~0.50 m. The top of the cave had two holes. One hole held a surveillance video camera (Sony CC-2) encased in an underwater Plexiglas housing (Ikelite 5210). The other hole held an underwater lamp that produced a dim red light that provided sufficient illumination for nocturnal recordings of the behaviour of lobsters inside the cave without disturbing the lobsters (Weiss *et al.*, 2006). Although this shelter design is rather crude, it provides important den requirements for *P. guttatus* [height, gripping surface, and darkness, see Lozano-Álvarez and Briones-Fourzán (2001)] while allowing the unhindered video-recording of all lobsters in its interior (Segura-García *et al.*, 2004; Weiss *et al.*, 2006). Lobsters were distinctively marked with strips of reflecting tape glued to the carapace to provide individual recognition.

The experiment consisted of four treatments that differed in social context and relative size of lobsters. The treatments denoted 2MF and 2FM consisted of mixed-sex groups (two males and two females of varied sizes), but in treatment 2MF the males were larger than the females, whereas in treatment 2FM one of the females was at least 5 mm carapace length (CL) larger than the largest male. The treatments denoted 4M and 4F were same-sex groups consisting of either four males (4M) or four females (4F) of varied sizes. Each treatment was replicated three times. The experiment was conducted over winter–spring of 2003, but due to the limited number of experimental units, the treatments were not run simultaneously (Table 1). To ensure that no lobster odours remained between treatments, at the end of each treatment all shelters were extracted from the tanks, brushed, rinsed, and sun dried, and the tanks were drained and thoroughly brushed.

Table 1. Dates and relevant characteristics of experimental periods for each treatment.

Treatment	Dates	Water temperature		
		Sunrise (h)	Sunset (h)	(°C, mean $\pm$ SD)
2MF	04–06 December	06:10	17:05	25.0 $\pm$ 1.9
4M	21–23 April	05:22	18:09	26.1 $\pm$ 1.0
4F	28–30 April	05:17	18:11	28.0 $\pm$ 0.6
2FM	09–11 June	06:05	19:29	27.8 $\pm$ 0.6

To avoid the potentially confounding effects of moult cycle and prior residence on aggressive behaviour (Atema and Cobb, 1980; Lipcius and Herrnkind, 1982; Peeke *et al.*, 1998), we used exclusively lobsters in intermoult as determined through microscopic observation of the tip of a pleopod (Lyle and MacDonald, 1983), and introduced the four lobsters in each group into the corresponding tank at the same time. All lobsters quickly took shelter in the cave, which was sufficiently large so as to loosely accommodate the entire group. Also, to avoid potential competition for food, which can occur among lobsters in captivity (Thomas *et al.*, 2003), lobsters were not fed for the duration of the experiment (72 h, see below). However, upon collection many lobsters were missing one or more limbs and several females were ovigerous (Table 2), as individual females of *P. guttatus* produce multiple clutches per year (Negrete-Soto *et al.*, 2002). Although inclusion of these factors was not contemplated originally, we included them *a posteriori* because aggressive behaviour can be affected by the lack of limbs (Juanes and Smith, 1995; Edsman and Jonsson, 1996) and, in females, by reproductive state (i.e. ovigerous vs. non-ovigerous; Figler *et al.*, 1995; Mello *et al.*, 1999).

After the lobsters acclimated to the experimental tanks for 24 h, the inside of the caves was continuously recorded for 72 h starting at 00:00 h of Day 1. A multiplexor (Panasonic WJ-FS216) allowed the simultaneous recording of the interior of the three caves on a time-lapse video recorder (Sony SVT-L400). We randomly chose 10 min from each hour of recording (one 5-min segment per each half-hour) to quantify the number of aggressions performed by each lobster (Martin and Bateson, 1993). Aggressions are defined as those offensive behaviours (i.e. attacks or threats) that elicit avoidance or escape behaviours in the opponent (Edwards and Herberholz, 2005; Table 3). All lobsters were used in one treatment only and were released back to the coral reef habitat at the end of the treatments.

Data analyses

We were interested in determining (i) the effect of social context (treatment) and time of day (i.e. night vs. day) on the per-capita rate of aggression (i.e. aggressions per hour per individual, Johnson *et al.*, 2004), (ii) whether individual traits (size, sex, number of missing limbs, and reproductive state) affected aggressiveness of individuals in a given social context, and (iii) whether irrespective of social context there was a significantly more aggressive individual in every group.

We did a factorial general linear model to examine the effects of social context (four levels corresponding to the four treatments) and time of day (two levels: day and night) on the per-capita rate of aggression (aggressions per individual per hour, Johnson *et al.*, 2004; response variable). We included time of day as a fixed factor because agonistic encounters tend to increase with the level of activity (Capelli and Hamilton, 1984), which for *P. guttatus* is substantially higher during the night (Lozano-Álvarez and Briones-Fourzán, 2001). For every treatment, we averaged the data from each unit across the night (from the hour of sunset to the hour of sunrise) and the day (from 1 h after sunrise to 1 h before sunset). Before analysis, the data were transformed to $\ln(\text{no. of aggressions} + 1)$ to increase homogeneity of variances as assessed with Levene's tests. For significant effects, differences in means were assessed with Tukey's HSD *post hoc* test for unequal sample sizes.

To examine whether individual traits (sex, size, number of missing limbs, and reproductive state) affected aggressiveness of lobsters in a given social context, we used a series of general

Table 2. Summary of data for lobsters used in experiment.

Treatment	Replicate unit	Lobster ID (sex and no.)	CL (mm)	No. of missing limbs	Reproductive state
2MF	A	M1	71.4	1	
		M2	69.0	0	
		F3	53.3	0	Non-ovigerous
		F4	50.0	1	Non-ovigerous
	B	M1	61.0	0	
		M2	58.0	2	
		F3	56.0	0	Non-ovigerous
		F4	54.5	1	Ovigerous
	C	M1	63.2	1	
		M2	59.7	0	
		F3	43.7	1	Non-ovigerous
		F4	40.8	1	Non-ovigerous
4M	A	M1	75.0	3	
		M2	61.7	1	
		M3	59.4	2	
		M4	43.0	3	
	B	M1	75.2	5	
		M2	60.7	1	
		M3	47.0	2	
		M4	40.9	0	
	C	M1	68.5	1	
		M2	64.7	3	
		M3	59.0	2	
4F	A	M1	51.4	1	
		F1	60.5	2	Non-ovigerous
		F2	52.3	1	Ovigerous
		F3	50.0	0	Ovigerous
	B	F4	42.8	4	Ovigerous
		F1	56.2	3	Non-ovigerous
		F2	55.0	0	Non-ovigerous
		F3	53.0	1	Non-ovigerous
	C	F4	49.0	2	Non-ovigerous
		F1	61.2	1	Ovigerous
		F2	47.3	1	Ovigerous
2FM	A	F3	45.8	2	Ovigerous
		F4	41.9	2	Non-ovigerous
		F1	55.9	0	Ovigerous
		M2	50.7	0	
	B	M3	45.0	0	
		F4	45.7	4	Ovigerous
		F1	60.4	1	Non-ovigerous
		M2	53.4	1	
	C	M3	53.0	0	
		F4	48.2	1	Ovigerous
		F1	55.0	2	Ovigerous
		M2	49.0	1	
		M3	44.0	1	
		F4	40.3	0	Non-ovigerous

M, male; F, female; CL, carapace length.

regression analyses. In all models, the dependent variable was the total number of aggressions performed by each individual throughout the experimental period, but the independent variables differed with social context. Thus, for the all-male context, we subjected the data from treatment 4M to a multiple regression analysis with male size and the number of missing limbs as continuous predictors. For the all-female context, we subjected the data from treatment 4F to an ANCOVA with reproductive state as a categorical factor (two levels: ovigerous and non-ovigerous), and female size and number of missing limbs as covariates. To examine the effect of sex in a

Table 3. Behaviours performed by lobsters involved in aggressive encounters [after Atema and Cobb (1980); Cobb (1981); Segura-García et al. (2004)].

Behaviour	Description
Attack or threat	
Antenna point	Direction of one or both antennae towards another lobster.
Approach	Movement towards another lobster (forwards, backwards, and sideways).
Clasp	To straddle another lobster and hug it using all pereopods.
Chase	Rapid approach towards a retreating animal.
Fight	Prolonged and intense aggressive contact.
Frontal approach	Two lobsters face to face in a high posture and with abdomens tucked.
Grasp	To reach out with pereopods, grab and hold onto part of another lobster.
Probe	Active probing or poking at another lobster with one or more pereopods.
Push	Body thrust at another lobster, resulting in its displacement.
Rush	Rapid approach, often with first two or three pairs of pereopods raised.
Escape or avoidance	
Avoid	Alteration of direction of motion, or reorientation of body because of an action by another lobster.
Crouch	To lower body so that ventral surface touches or almost touches the substrate.
Flee	Rapid movement away from another lobster.
Tail flip	Backward withdrawal by the abdomen flex escape response.
Walk away	Retreat from another lobster (backward, sideward, or forward).

mixed-sex context, we pooled the data from treatments 2MF and 2FM. Pooling the data from different treatments is not warranted, but we did this to increase the sample size to do an ANCOVA with sex as a categorical factor and size and the number of missing limbs as covariates. Before all analyses, the size data were ln-transformed, and the dependent variable was transformed to ln (no. of aggressions + 1) to increase homogeneity of variances.

Finally, we used goodness-of-fit χ^2 analyses to test the null hypothesis that all four lobsters in each experimental unit of a given treatment were equally aggressive. If the test was significant, we then proceeded with subdivided χ^2 analyses (Zar, 1999) to determine whether or not there was a significantly more aggressive individual in that unit and interpreted the results based on individual traits and the corresponding social context.

Results

Time of day and aggressiveness

The per-capita rate of aggression (mean $\pm$ SE) varied significantly with time of day (night: 1.05 ± 0.11 aggressions h^{-1} ; day: 0.53 ± 0.06 aggressions h^{-1} ; $F_{1, 226} = 20.456$, $p < 0.001$) and treatment, with higher values in treatments 4M and 2FM (1.03 ± 0.12 and 0.98 ± 0.15 aggressions h^{-1} , respectively) than in treatments 4F and 2MF (0.51 ± 0.08 and 0.42 ± 0.09 aggressions h^{-1} , respectively; $F_{3, 226} = 7.984$, $p < 0.001$). However, the interaction term was also significant ($F_{3, 226} = 4.204$, $p = 0.006$), indicating that the effect of time of day on aggressiveness varied with social context. Indeed, only in treatments 2MF and 2FM was the per-capita rate of aggression

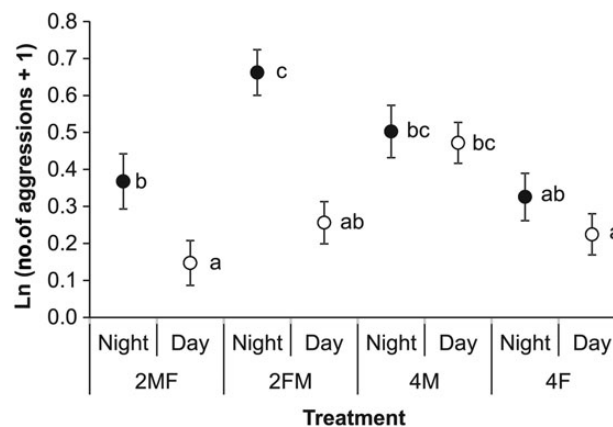


Figure 1. Night (filled circles) vs. day (open circles) per-capita rate of aggression (no. of aggressions per individual per hour) in groups of four *P. guttatus* held in different social contexts (treatments): 2MF, two males and two females and the males were larger than the females; 2FM, two males and two females but the largest individual was a female; 4M, four males of different sizes; 4F: four females of different sizes. Error bars denote 95% confidence interval. Different letters indicate statistically different groups (treatment $\times$ day/night interaction).

significantly higher at night than during the day (Figure 1). In treatments 4M and 4F, in contrast, the rate of aggression did not differ significantly between night and day, but was higher for both periods in treatment 4M relative to treatment 4F (Figure 1).

Effects of individual traits and social context on aggressive behaviour

The regression model for the data from lobsters in an all-male context (treatment 4M) was significant ($R^2_{adj} = 0.449$; $F_{2, 9} = 5.484$, $p = 0.028$). Partial correlations revealed that the number of aggressions performed by individual males increased with size ($b = 0.626$, $SE = 0.253$, $t = 2.477$, $p = 0.035$; Figure 2a), but decreased with an increasing number of missing limbs ($b = -0.783$, $SE = 0.253$, $t = 3.097$, $p = 0.013$; Figure 2b). In contrast, the regression model for the data from lobsters in an all-female context (treatment 4F) was not significant ($R^2_{adj} = 0.327$; $F_{3, 8} = 2.780$, $p = 0.110$), although partial correlations showed that the number of aggressions performed by individual females also increased with size ($b = 0.685$, $SE = 0.273$, $t = 2.507$, $p = 0.036$; Figure 2c), but had no apparent relationship with either the number of missing limbs ($b = -0.091$, $SE = 0.267$, $t = -0.341$, $p = 0.741$; Figure 2d) or maternal condition ($b = -0.042$, $SE = 0.257$, $t = -0.162$, $p = 0.875$).

The regression model for the data from the mixed-sex context was significant ($R^2_{adj} = 0.290$; $F_{3, 20} = 4.134$, $p = 0.0197$). Partial correlations showed that neither the number of missing limbs ($b = 0.137$, $SE = 0.180$, $t = 0.761$, $p = 0.455$) nor size ($b = 0.047$, $SE = 0.190$, $t = 0.246$, $p = 0.808$) had a significant effect on the number of aggressions, which was expected due to the difference in relative size of males and females between treatments 2MF and 2FM. However, sex did have a significant effect on the number of aggressions ($b = 0.611$, $SE = 0.191$, $t = 3.204$, $p = 0.004$), with males performing, on average, four times as many aggressions (15.92 ± 4.79 aggressions lobster $^{-1}$) as females (3.67 ± 1.09 aggressions lobster $^{-1}$).

Was there a more aggressive individual in each group?

In all three replicate units (denoted A, B, and C) of all treatments, the number of aggressions varied significantly among individuals

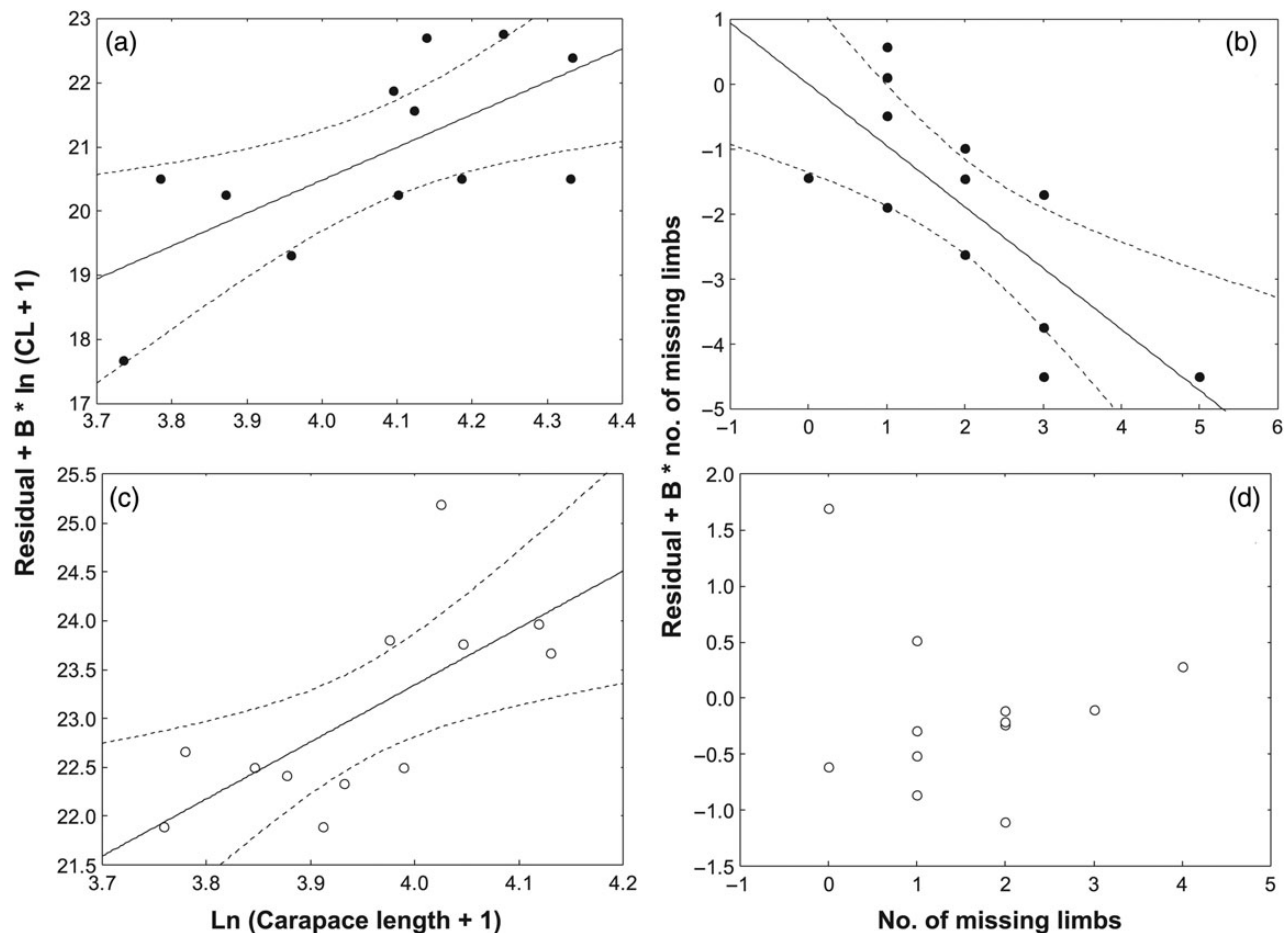


Figure 2. Partial residual plots for (a and c) carapace length, CL (ln-transformed) and (b and d) number of missing limbs vs. total number of aggressions performed by (a and b) individual males of *P. guttatus* held in an all-male context (treatment 4M, black dots), and (c and d) individual females held in an all-female context (treatment 4F, white dots). Dotted lines are 95% confidence interval. Plots for treatment 4M were obtained from a general regression model (GRM) evaluating the effects of both covariates on the number of aggressions performed by each male. Plots for treatment 4F were obtained from a GRM evaluating the effects of size, number of missing limbs, and reproductive state on the total number of aggressions performed by each female. The effects on the number of missing limbs and reproductive state were not significant.

(χ^2 tests, all p -values < 0.001), except in unit A of treatment 4F ($\chi^2_3 = 3.514$, $p = 0.173$). Subdivided χ^2 analyses revealed that, in treatments 2MF and 4M, there was a clearly more aggressive individual in each unit, whereas in treatments 4F and 2FM there was wide variability among units. In all units of treatment 2MF, the most aggressive individual was the largest male (Figure 3a). In treatment 4M, it was the largest male in unit C, but the second largest male in units A and B (Figure 3b). However, the second largest males in units A and B were missing one limb each, whereas the largest males were missing three and five limbs, respectively, and were not observed performing aggressions (Figure 3b). In treatment 4F, all four females in unit A exhibited a similar, relatively low level of aggressiveness, but the most aggressive individual in unit C was the largest female and in unit B it was the second largest female (Figure 3c). In treatment 2FM, despite the largest individual in all units being a female, the more aggressive individual in unit A was the largest male, whereas the two males in unit B exhibited a similar level of aggressiveness, higher than that of both females. In contrast, in unit C, the two males and the largest female performed a similar number of aggressions (Figure 3d).

Discussion

Our results provide further insights into the shelter-related aggressive behaviour of *P. guttatus* lobsters in different social contexts. Aggressions were generally of low intensity, with only occasional bouts of strong intensity. Males exhibited a higher level of aggressiveness than females, but the number of missing limbs significantly impacted the degree of aggressiveness in males, and size was an important predictor of aggressiveness among individuals of the same sex. Because the treatments were not run concurrently, there is some possibility that the results may be related with the sequence in which the treatments were run. However, [Lipcius and Herrnkind \(1987\)](#) found that photoperiod and temperature did not significantly influence levels of aggression in *Panulirus argus*, a species that co-occurs with *P. guttatus*, probably because aggression is expressed in many activities that tropical spiny lobsters perform year-round (e.g. territorial disputes and reproductive activities; [Berry, 1970](#); [Lipcius and Herrnkind, 1982](#)).

Consistent with previous studies on *P. guttatus* ([Lozano-Álvarez and Briones-Fourzán, 2001](#); [Segura-García et al., 2004](#)), most aggressions were related to the occupancy of the shelter space

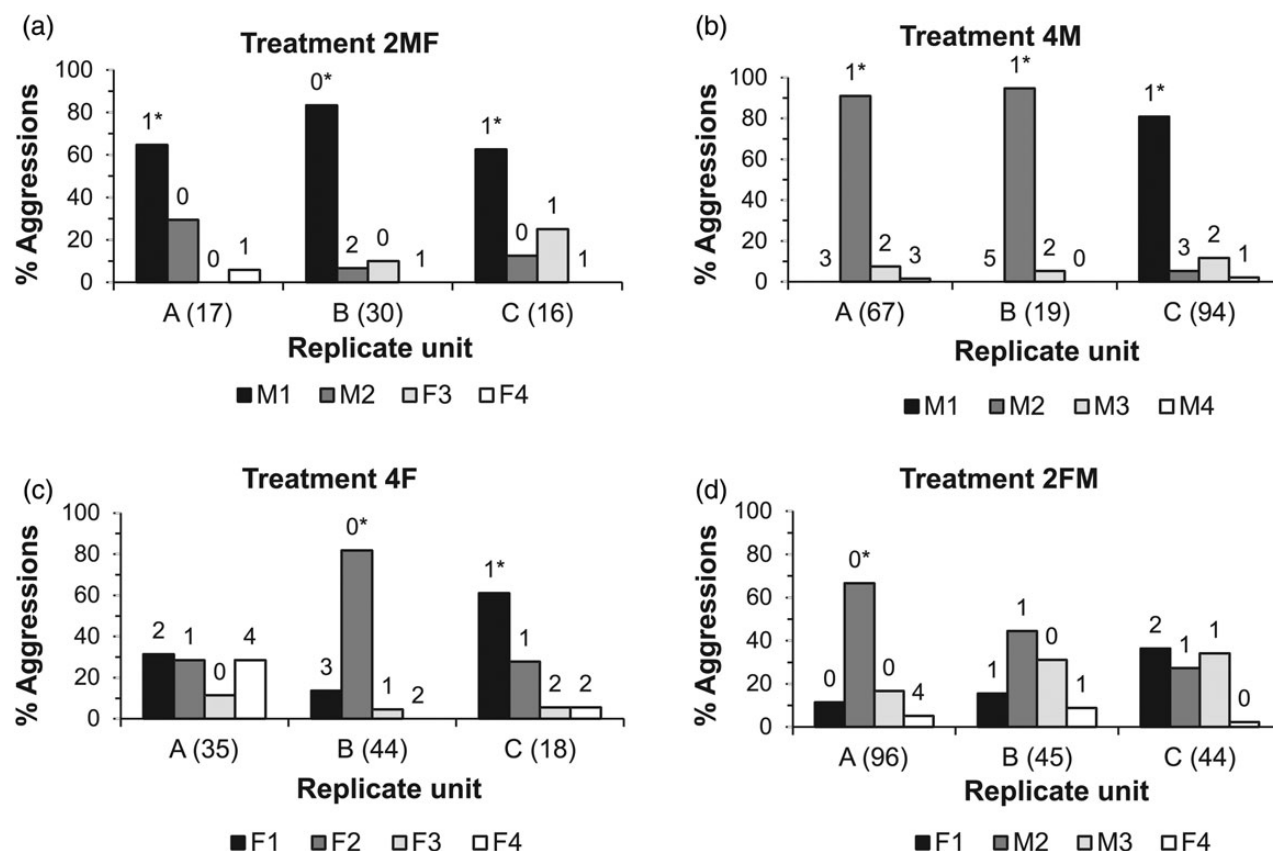


Figure 3. Percentage of the total number of aggressions performed by each lobster ($n = 4$) in each replicate unit (a, b and c) per treatment. Numbers in parentheses on the x-axes are the total number of aggressions recorded in that unit. Numbers above columns denote the number of limbs that each lobster was missing. In the graph legends, F is female, M is male, and the numbers 1–4 indicate the size of lobsters in a decreasing order (i.e. 1 is largest and 4 is smallest). Asterisks denote individuals that performed significantly more aggressions in a given unit.

(with the vault tending to be the preferred spot), but tended to subside when individuals were sufficiently spaced out. Spacing is an important mechanism to avoid future agonistic interactions in other decapods (Clayton, 1988; Fero and Moore, 2008; Daws et al., 2011) and possibly also in *P. guttatus*. Spacing could also reflect the dependence of *P. guttatus* lobsters on cryptic behaviour as a mechanism to avoid detection by predators, as opposed to the communal defensive behaviour exhibited by other, more social spiny lobsters such as *P. argus* (Briones-Fourzán et al., 2006; Lozano-Álvarez et al., 2007).

We expected the per-capita rate of aggression to be greater at night but this only occurred in the mixed-sex groups, although less markedly in treatment 2MF (wherein the two largest individuals were males) relative to treatment 2FM (wherein the largest individual was a female). This result likely reflects the occurrence of a clearly more aggressive individual (the largest male) in each replicate unit of treatment 2MF. In many decapods, once the dominant individual in a group asserts itself, aggressive encounters tend to become fewer and less intense because dominants and subordinates are identified through status-related cues (Capelli and Hamilton, 1984; Huber and Kravitz, 1995; Karavanich and Atema, 1998; Edwards and Herberholz, 2005). In an experiment with *P. argus*, for example, once some individuals became dominants and others subordinates, urine-borne signals released by the dominants induced avoidance behaviour from subordinates, reducing the general level of aggression (Shabani et al., 2009). In treatment 2FM, in contrast, the

per-capita rate of aggression was higher and there was great variability in the number of aggressions within and among replicate units. These findings suggest that when the largest individuals in mixed-sex groups are males, the largest male will probably assert itself rather quickly, reducing the general level of aggression in the group, but when the largest individual is a female, it may take longer for any one lobster to assert itself.

The per-capita rate of aggression did not vary significantly with time of day in the same-sex treatments, but was twice as high in treatment 4M as in treatment 4F, reflecting the generally higher degree of aggressiveness in males. Size was an important predictor of aggressiveness in both these treatments, indicating that, in the absence of males, large females can also behave aggressively towards smaller females, as has been observed in other decapods (Lee and Fielder, 1983; Thorpe et al., 1994; Edsman and Jonsson, 1996). In crayfish *Procambarus clarkii* and clawed lobsters *Homarus americanus*, ovigerous females are more aggressive than non-ovigerous females, presumably to ensure offspring survival (Figler et al., 1995; Mello et al., 1999), but reproductive state did not affect aggressiveness in females of *P. guttatus*. However, as ovigerous and non-ovigerous females were not equally present in all replicates of treatment 4F, this result should be considered as preliminary.

Antennal contact is important in the social behaviour of many decapods (Solon and Cobb, 1980; Mello et al., 1999; Vickery et al., 2012), but its relative importance in aggressive encounters varies with taxa. For example, crayfish and clawed lobsters use their

antennae to whip and attack conspecifics (Solon and Cobb, 1980; Edsman and Jonsson, 1996; Smith and Dunham, 1996), whereas spiny lobsters use their antennae to threaten rather than to attack conspecifics (Atema and Cobb, 1980; Cobb, 1981; Shabani *et al.*, 2009). We observed that the rapid pointing of antennae by a large lobster towards a smaller conspecific occasionally sufficed to elicit the retreat of the latter.

In decapods, the degree to which individual performance in agonistic encounters is affected by the lack of limbs varies with the type and number of missing limbs (Neil, 1985; Juanes and Smith, 1995). In the clawless spiny lobsters, agonistic interactions largely involve the use of pereopods (Table 3). This, in conjunction with a higher level of aggressiveness of males of *P. guttatus* relative to females, could explain why, in an all-male context, aggressiveness decreased with an increasing number of missing limbs. In male spiny lobsters, the first three pairs of pereopods are longer and thicker than in females and play an important role in agonistic and mating behaviour (McKoy, 1979; Lipcius *et al.*, 1983). Therefore, large intact males would have a competitive advantage over smaller males or males lacking limbs. However, individuals that lose limbs may benefit from becoming subordinate by avoiding costs such as increased energy expenditure, further injury from a conspecific, or increased predation risk (Edsman and Jonsson, 1996; Fleming *et al.*, 2007). For example, Berry (1970) observed that the two largest males in a group of *Panulirus homarus* lobsters sharing a tank were missing several limbs and were subordinate to a smaller but intact male. We found that, in an all-male context, aggressiveness of males decreased with an increasing number of missing limbs to the point of neutralizing the advantage conferred by a large size. Thus, the most aggressive individual in all replicate units of treatment 4M was the male that had fewer missing limbs in conjunction with a relatively large size.

In females, the apparent lack of relationship between the number of missing limbs and aggressiveness must be taken with caution because, unlike in treatment 4M wherein one-third of the test males were severely handicapped (i.e. missed three limbs or more) and these males were present in all three replicates, in treatment 4F only 2 of the 12 test females were severely handicapped and no such females were present in one replicate. In addition, reproductive condition of females, which varied widely within and between replicates, may have interacted in complex ways with the number of missing limbs of individual females, obscuring the relative importance of these two factors.

Our results also provide insights into some potential consequences of the loss of habitat complexity and increasing fishing pressure for these obligate reef-dwelling lobsters. Density of *P. guttatus* increases with habitat quality (i.e. coral reef complexity), because complex reefs provide a wide variety of refuges and foraging spaces (Wynne and Côté, 2007; Robertson and Butler, 2009). However, Caribbean coral reefs are losing structural complexity and becoming increasingly flatter, due in part to the loss of key reef-building, morphologically complex corals that are being replaced by faster-growing species that are more resistant to stress but structurally simpler (Álvarez-Filip *et al.*, 2011). The tendency of *P. guttatus* lobsters to space out in separate cavities within larger crevices presumably reduces agonistic interactions and detection by predators. Therefore, habitat constraints and loss of complexity could increase local levels of aggressiveness of these lobsters, as has been observed for other species (Basquill and Grant, 1998; Corkum and Cronin, 2004), particularly in periods of high reproductive activity when their tendency to aggregate increases (Briones-Fourzán *et al.*, 2013).

In Caribbean locations where *P. guttatus* is heavily exploited, fishing pressure can reduce the mean size of lobsters and even alter sex ratios by disproportionately extracting the largest individuals, which are mostly males (Evans *et al.*, 1996; Wynne and Côté, 2007). Because large females of *P. guttatus* prefer to mate with larger males (Magallón-Gayón *et al.*, 2011) and our results suggest that, in the absence of large males, it can take longer for smaller males to assert themselves, fishing pressure may also indirectly lead to an increased level of aggressiveness among these spiny lobsters.

Acknowledgements

We greatly appreciate the highly skilled technical support in field and laboratory activities provided by F. Negrete-Soto and C. Barradas-Ortiz. Additional help was provided by A. Osorio-Arciniegas, E. Ramírez-Zaldívar, M. Pérez-Ortiz, and J. Valladárez-Cob. Three anonymous reviewers greatly improved the manuscript. This study was funded by Universidad Nacional Autónoma de México and Consejo Nacional de Ciencia y Tecnología (Mexico) through grant 40159-Q and a B.Sc. scholarship for RDG. Annual permits to capture lobsters for experimental purposes were issued by Comisión Nacional de Acuacultura y Pesca. Experiments complied with the current laws of Mexico.

References

- Álvarez-Filip, L., Dulvy, N. K., Côté, I. M., Watkinson, A. R., and Gill, J. A. 2011. Coral identity underpins reef complexity on Caribbean reefs. *Ecological Applications*, 21: 2223–2231.
- Atema, J., and Cobb, J. S. 1980. Social behavior. In *The Biology and Management of Lobsters*, Vol. 1, pp. 409–450. Ed. by J. S. Cobb, and B. F. Phillips. Academic Press, New York. 463 pp.
- Basquill, S. P., and Grant, J. W. A. 1998. An increase in habitat complexity reduces aggression and monopolization of food by zebra fish (*Danio rerio*). *Canadian Journal of Zoology*, 76: 770–772.
- Beattie, C. L., Pitt, K. A., and Connolly, R. M. 2012. Both size and gender of mud crabs influence the outcomes of interference interactions. *Journal of Experimental Marine Biology and Ecology*, 434–435: 1–6.
- Berrill, M. 1976. Aggressive behavior of post-plerulus larvae of the western rock lobster *Panulirus longipes* (Milne-Edwards). *Australian Journal of Marine and Freshwater Research*, 27: 83–88.
- Berry, P. F. 1970. Mating behavior, oviposition and fertilization in the spiny lobster *Panulirus homarus* (Linnaeus). *Oceanographic Research Institution*, (Durban), Investigational Report 24: 1–16.
- Briones-Fourzán, P., and Lozano-Álvarez, E. 2005. Seasonal variations in chemical response to conspecific scents in the spotted spiny lobster, *Panulirus guttatus* (Latreille). *New Zealand Journal of Marine and Freshwater Research*, 39: 383–390.
- Briones-Fourzán, P., Magallón-Gayón, E., and Lozano-Álvarez, E. 2013. Increased reproductive opportunity: a potential benefit of seasonal aggregation for a little-gregarious and highly sedentary spiny lobster. *Marine Biology Research*, 9: 77–87.
- Briones-Fourzán, P., Pérez-Ortiz, M., and Lozano-Álvarez, E. 2006. Defense mechanisms and anti-predator behavior in two sympatric species of spiny lobsters, *Panulirus argus* and *P. guttatus*. *Marine Biology*, 149: 227–239.
- Capelli, G. M., and Hamilton, P. A. 1984. Effects of food and shelter on aggressive activity in the crayfish *Orconectes rusticus* (Girard). *Journal of Crustacean Biology*, 4: 252–260.
- Childress, M. J. 2007. Comparative sociobiology of spiny lobsters. In *Evolutionary Ecology of Social and Sexual Systems: Crustaceans as Model Organisms*, pp. 271–293. Ed. by J. M. Duffy, and M. Thiel. Oxford University Press, Oxford. 520 pp.
- Clayton, D. A. 1988. Hood construction as a spacing mechanism in *Cleistostoma kuwaitense* (Crustacea: Ocypodidae). *Marine Biology*, 99: 57–61.

- Cobb, J. S. 1981. Behaviour of the Western Australian spiny lobster, *Panulirus cygnus* George, in the field and laboratory. Australian Journal of Marine and Freshwater Research, 32: 399–409.
- Corkum, L. D., and Cronin, D. J. 2004. Habitat complexity reduces aggression and enhances consumption in crayfish. Journal of Ethology, 22: 23–27.
- Davis, K. M., and Huber, R. 2007. Activity patterns, behavioural repertoires, and agonistic interactions of crayfish: a non-manipulative field study. Behaviour, 144: 229–247.
- Daws, A. G., Hock, K., and Huber, R. 2011. Spatial structure of hierarchical groups: testing for processes of aggregation, clustering, and spatial centrality in crayfish (*Orconectes rusticus*). Marine and Freshwater Behaviour and Physiology, 44: 209–222.
- Drews, C. 1993. The concept and definition of dominance in animal behavior. Behaviour, 125: 283–313.
- Edsman, L., and Jonsson, A. 1996. The effect of size, antennal injury, ownership, and ownership duration on fighting success in male signal crayfish, *Pacifastacus leniusculus* (Dana). Nordic Journal of Freshwater Research, 72: 80–87.
- Edwards, D. H., and Herberholz, J. 2005. Crustacean models of aggression. In *The Biology of Aggression*, pp. 38–61. Ed. by R. J. Nelson. Oxford University Press, Oxford. 528 pp.
- Evans, C. R., Lockwood, A. P. M., and Evans, A. J. 1996. Field studies of the population dynamics of the spotted spiny lobster *Panulirus guttatus* (Latreille) at Bermuda. Gulf of Mexico Science, 14: 55–65.
- Fero, K., and Moore, P. A. 2008. Social spacing of crayfish in natural habitats: what role does dominance play? Behavioral Ecology and Sociobiology, 62: 1119–1125.
- Fielder, D. R. 1965. A dominance order for shelter in the spiny lobster *Jasus lalandei* (H. Milne-Edwards). Behaviour, 24: 236–245.
- Figler, M. H., Twum, M., Finkelstein, J. E., and Peeke, H. V. S. 1995. Maternal aggression in red swamp crayfish *Procambarus clarkii* Girard: the relation between reproductive status and outcome of aggressive encounters with male and female conspecifics. Behaviour, 132: 107–125.
- Fleming, P. A., Muller, D., and Bateman, P. W. 2007. Leave it all behind: a taxonomic perspective of autotomy in invertebrates. Biological Reviews, 82: 481–510.
- Graham, M. E., and Herberholz, J. 2009. Stability of dominance relationships in crayfish depends on social context. Animal Behaviour, 77: 195–199.
- Hand, J. L. 1986. Resolution of social conflicts: dominance, egalitarianism, spheres of dominance, and game theory. Quarterly Review of Biology, 61: 201–220.
- Huber, R., and Kravitz, E. 1995. A quantitative analysis of agonistic behavior in juvenile American lobsters (*Homarus americanus*). Brain, Behavior and Evolution, 46: 72–83.
- Huber, R., and Kravitz, E. 2010. Aggression: towards an integration of gene, brain and behaviour. In *Social Behaviour: Genes, Ecology and Evolution*, pp. 165–180. Ed. by T. Székely, A. J. Moore, and J. Komdeur. Cambridge University Press, Cambridge. 576 pp.
- Johnson, C. A., Grant, J. W. A., and Giraldeau, L. A. 2004. The effect of patch size and competitor number on aggression among foraging house sparrows. Behavioral Ecology, 15: 412–418.
- Juanes, F., and Smith, L. D. 1995. The ecological consequences of limb damage and loss in decapod crustaceans: a review and prospectus. Journal of Experimental Marine Biology and Ecology, 193: 197–223.
- Karavanich, C., and Atema, J. 1998. Individual recognition and memory in lobster dominance. Animal Behaviour, 56: 1553–1560.
- Karnofsky, E. B., Atema, J., and Elgin, R. H. 1989. Field observations of social behavior, shelter use, and foraging in the lobster, *Homarus americanus*. Biological Bulletin, 176: 239–246.
- Lee, C. L., and Fielder, D. R. 1983. Agonistic behavior and the development of dominance hierarchies in the freshwater prawn *Macrobrachium australiense* Holthuis 1950 (Crustacea: Palaemonidae). Behaviour, 83: 1–17.
- Lipcius, R. N., Edwards, M. L., Herrnkind, W. F., and Waterman, S. A. 1983. In situ mating behavior of the spiny lobster *Panulirus argus*. Journal of Crustacean Biology, 3: 217–222.
- Lipcius, R. N., and Herrnkind, W. F. 1982. Molt cycle alterations in behavior, feeding, and diel rhythms of a decapod crustacean, the spiny lobster *Panulirus argus*. Marine Biology, 68: 241–252.
- Lipcius, R. N., and Herrnkind, W. F. 1987. Control and coordination of reproduction and molting in the spiny lobster *Panulirus argus*. Marine Biology, 96: 207–214.
- Lozano-Álvarez, E., and Briones-Fourzán, P. 2001. Den choice and occupation patterns of shelters in two sympatric lobster species, *Panulirus argus* and *Panulirus guttatus*, under experimental conditions. Marine and Freshwater Research, 52: 1145–1155.
- Lozano-Álvarez, E., Briones-Fourzán, P., Osorio-Arciniegas, A., Negrete-Soto, F., and Barradas-Ortiz, C. 2007. Coexistence of congeneric spiny lobsters on coral reefs: differential use of shelter resources and vulnerability to predators. Coral Reefs, 26: 361–373.
- Lozano-Álvarez, E., Carrasco-Zanini, G., and Briones-Fourzán, P. 2002. Homing and orientation in the spotted spiny lobster, *Panulirus guttatus* (Decapoda: Palinuridae), towards a subtidal coral reef habitat. Crustaceana, 75: 859–873.
- Lyle, W. G., and MacDonald, C. D. 1983. Molt stage determination in the Hawaiian spiny lobster *Panulirus marginatus*. Journal of Crustacean Biology, 3: 208–216.
- Magallón-Gayón, E., Briones-Fourzán, P., and Lozano-Álvarez, E. 2011. Does size always matter? Mate choice and sperm allocation in *Panulirus guttatus*, a highly sedentary, habitat-specialist spiny lobster. Behaviour, 148: 1333–1358.
- Martin, P., and Bateson, P. 1993. Measuring Behaviour: An Introductory Guide, 2nd edn. Cambridge University Press, Cambridge. 222 pp.
- McKoy, J. L. 1979. Mating behaviour and egg laying in captive rock lobster, *Jasus edwardsii* (Crustacea: Decapoda: Palinuridae). New Zealand Journal of Marine and Freshwater Research, 13: 407–413.
- Mello, J. J., Cromarty, S. I., and Kass-Simon, G. 1999. Increased aggressiveness in gravid American lobsters, *Homarus americanus*. Aggressive Behaviour, 25: 451–472.
- Negrete-Soto, F., Lozano-Álvarez, E., and Briones-Fourzán, P. 2002. Population dynamics of the spiny lobster *Panulirus guttatus* in a coral reef on the Mexican Caribbean. Journal of Shellfish Research, 21: 279–288.
- Neil, S. J. 1985. Size assessment and cues: studies of hermit crab contests. Behaviour, 92: 22–38.
- Peeke, H. V. S., Figler, M. H., and Chang, E. S. 1998. Sex differences and prior residence effects in shelter competition in juvenile lobsters, *Homarus americanus* Milne-Edwards. Journal of Experimental Marine Biology and Ecology, 229: 149–156.
- Robertson, D. N., and Butler, M. J. 2009. Variable reproductive success in fragmented populations. Journal of Experimental Marine Biology and Ecology, 377: 84–92.
- Segura-García, I., Lozano-Álvarez, E., and Briones-Fourzán, P. 2004. Within-shelter behaviour of the spotted spiny lobster, *Panulirus guttatus* (Latreille), in simulated communal dens: an exploratory study. Marine and Freshwater Behaviour and Physiology, 37: 17–30.
- Shabani, S., Kamio, M., and Derby, C. D. 2009. Spiny lobsters use urine-borne olfactory signaling and physical aggressive behaviors to influence social status of conspecifics. Journal of Experimental Biology, 212: 2464–2474.
- Sharp, W. C., Hunt, J. H., and Lyons, W. G. 1997. Life history of the spotted spiny lobster, *Panulirus guttatus*, an obligate reef-dweller. Marine and Freshwater Research, 48: 687–698.
- Smith, M. R., and Dunham, D. W. 1996. Antennae mediate agonistic physical contact in the crayfish *Orconectes rusticus* (Girard, 1852) (Decapoda, Cambaridae). Crustaceana, 69: 668–674.
- Solon, M. H., and Cobb, J. S. 1980. Antennae whipping behavior in the American lobster *Homarus americanus* (Milne-Edwards). Journal of Experimental Marine Biology and Ecology, 48: 217–224.

- Thomas, C. W., Carter, C. G., and Crear, B. J. 2003. Feed availability and its relationship to survival, growth, dominance and the agonistic behavior of the southern rock lobster, *Jasus edwardsii* in captivity. *Aquaculture*, 215: 45–65.
- Thorpe, K., Huntingford, F., and Taylor, A. 1994. Relative size and agonistic behavior in the female velvet swimming crab, *Necora puber* (L.) (Brachyura, Portunidae). *Behavioral Processes*, 32: 235–246.
- Vickery, R., Hollowell, K., and Hughes, M. 2012. Why have long antennae? Exploring the function of antennal contact in snapping shrimp. *Marine and Freshwater Behaviour and Physiology*, 45: 161–176.
- Weiss, H. M., Lozano-Álvarez, E., Briones-Fourzán, P., and Negrete-Soto, F. 2006. Using red light with fixed-site video cameras to study the behavior of spiny lobsters, *Panulirus argus*, and associated animals at night and inside their shelters. *Marine Technology Society Journal*, 40: 86–95.
- Wynne, S. P., and Côté, I. M. 2007. Effects of habitat quality and fishing on Caribbean spotted spiny lobster populations. *Journal of Applied Ecology*, 44: 488–494.
- Zar, J. H. 1999. *Biostatistical Analysis*, 4th edn. Prentice-Hall, Upper Saddle River, NJ. 929 pp.

Handling editor: Howard Browman